# #ManyDaughters: Pre-analysis plan

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This document outlines the pre-analysis plan (PAP) for testing the four hypotheses proposed in the #ManyDaughters project, using data from the German Socio-Economic Panel (SOEP). We begin by addressing analytical issues that are common across all four hypotheses and detailing our strategies for mitigating them. Subsequently, for each hypothesis, we provide a description of the variables to be used, the inclusion criteria, the analytical approach, and the way in which the results will be interpreted to assess the support for each hypothesis.

## Common Issues in the Proposed Hypotheses

### *Endogeneity*

Comparing the attitudes and behaviors of parents with daughters to those without introduces three key endogeneity concerns. First, the probability of having a daughter is inherently tied to the initial decision to have children—a decision that may itself be shaped by pre-existing attitudes about gender roles and political preferences. Second, individuals who opt to have more children are statistically more likely to have at least one daughter; and the choice to have a larger family may also reflect underlying gender-equality attitudes or political preferences. Third, attitudes regarding gender equality or political preferences could influence the desired gender composition of offspring, potentially leading to different stopping rules in family planning.

### *Exposure*

Testing the four hypotheses requires a clear specification of the time frame within which the effect of having a daughter is expected to manifest. Specifically, we must consider whether the presence of a daughter leads to an immediate shift in gender-equality attitudes, behaviors, and political preferences, or whether such changes emerge only after a period of sustained parental exposure. In this analysis, we adopt the assumption that at least *one year* must elapse before any measurable influence of having a daughter on parental attitudes and behaviors can be expected to occur.

### *Solution: A restricted sample*

To address concerns related to endogeneity and exposure, and to minimize potential bias, we adopt a conservative sampling strategy. Specifically, we restrict our analysis to *parents with only one biological child who is at least one year old*. Limiting the sample to one-child families helps avoid the confounding issue of comparing parents of daughters with non-parents, since the decision to become a parent may itself be influenced by pre-existing gender attitudes or political preferences. Additionally, excluding families with multiple children mitigates two further concerns: first, that gender-related or political attitudes may influence the decision to have larger families; and second, that such attitudes could shape the preferred gender composition of children through selective stopping behavior.

While this approach does not allow us to examine the effects of having more than one daughter, it represents a deliberately conservative strategy. If the proposed influences of having daughters are indeed true, we expect their effects to be observable within this restricted sample.

**Hypothesis 1 (H1). Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for respondents with daughters) > (X for respondents without daughters).**

***Variables***

To test Hypothesis 1 (H1), we adopt the definition of gender equality provided by the European Institute for Gender Equality (EIGE), an agency of the European Union. According to EIGE, gender equality refers to a state in which women and men and girls and boys have equal rights, responsibilities and opportunities.[1] While the SOEP dataset does not include a single variable that directly captures attitudes aligned with this definition, we draw on the following set of SOEP variables that reflect its core dimensions.

**plj0620**: democracy: women have the same rights as men (available for 2016, 2017, and 2018).
**plj0624**: position of women: education of son more important than of daughter (2016, 2017, 2018).
**plj0625**: position of women: husband should have the final say (2016, 2017, 2018).
**lb1120**: Fam: Man/Women Equal in Household/Family and Job (available for all years in SOEP-core).
**lb1113**: Fam: Women Should Rather Care About Family Than Career (SOEP-core).

We view gender equality as a latent construct—unobservable directly but inferred through related variables. We assume that the five previously identified observable variables are correlated with this latent dimension. Based on this assumption, we conduct Principal Component Analysis (PCA) to extract the dominant components and construct a composite index of gender equality. The resulting score is rescaled to range from 0 to 1. We denote this composite index as *EG*, which serves as the dependent variable in the subsequent regression analysis.

The biological child is identified with variable **lb1178**.

***Inclusion Criteria***

The observations to be analyzed include parents with only one biological child who is at least one year old. Since the variables used to construct *EG* are available only for the years 2016 to 2018, our analysis is limited to this period. The sample includes all respondents regardless of civil status, encompassing single, married, and divorced individuals.

[1] https://eige.europa.eu/publications-resources/thesaurus/terms/1059?language_content_entity=en#:~:text=Gender%20equality%20implies%20that%20the,men%20as%20well%20as%20women

*Exact Analysis*

We will first conduct balance diagnostics of the group of parents of a daughter and the group of parents of a son. Specifically, we will compare the two groups in terms of observable characteristics that could influence responders' attitudes toward gender equality.

**Age**: Responder's age to control for differences in gender equality across age.
**Gender**: Responders' gender to account for differences between men and women in their gender equality preferences.
**Education** (plg0268): Categorical variable indicating the completion of a degree, including secondary school-leaving certificate, university degree, and vocational degree; used to account for the effect of education in driving gender equality preferences.
**Number of sisters** (lb0063): Count of sisters reported by the responder, used to control for exposure to women during the respondents' childhood at their homes.
**Religion preferences** (plh0258_v10, plh0258_v11): Categorical variables revealing distinct region affiliations. The former describes affiliation with Christian organizations, while the second describes affiliation with Islamic organizations. Differences in religious affiliations could determine responders' beliefs and attitudes toward gender equality.
**Employment status** (plb0022_h): Categorical variable indicating the employment status of the responders, including employed full-time, not employed, doing internship, completing in-service training, marginally or irregularly employed, in partial retirement, phase with zero working hours. This variable is used to account for the influence of current employment status in individuals' attitudes to gender equality.
**Citizenship** (plj0022 and plj0023_v1): The first variable classifies responders depending on whether they have or do not have German citizenship, whereas the second identifies any additional citizenship the responders may have. To account for the distinct cultural background influence in driving gender equality preferences.
**Salary previous year** (plb0471_h): Gross salary in euros of the prior year. To take into account the income influence.

We will conduct mean comparison tests between parents of daughters and parents of sons for each of the variables listed above. Where statistically significant differences are identified, we will include those variables as controls in the subsequent regression analyses to account for potential confounding factors.

To test H1, we will use a regression model estimated by Ordinary Least Squares (OLS). The model specification is:

$$EG_i = \beta_0 + \beta_1 Daughter_i + \boldsymbol{W}\delta + c_y + \varepsilon_i \quad (1)$$

Where $EG$ is our composite measuring gender equality, $Daughter$ is a binary variable taking the value of one if the respondent has a daughter and zero if she has a son. We will use the SOEP variable, lb0288_v2, to define the independent variable $Daughter$. We will recodify lb0288_v2 so that one represents a daughter and zero represents a son. $\boldsymbol{W}$ is a vector of control variables, while $c_y$ is a set of fixed effects for the age of the first-born child of the responder. $\boldsymbol{W}$ includes controls that help balance the sample. Accounting for $c_y$ allows us to consider the exposition

effect that having a daughter may have on the responders' attitudes toward gender equality, compared to having a son.

This model is estimated using Huber-White standard errors to mitigate the impact of observing heteroskedasticity.

***Interpretation***

The parameter estimate of interest is $\beta_1$, which measures differences between responders' parenting a daughter and those parenting a son in their support for gender equality. Hence, $\beta_1$ measures the relative impact of having a daughter, compared to a son, on the respondents' support for gender equality. H1 receives support when $\hat{\beta}_1 > 0$, and p-value < .05.

**Hypothesis 2 (H2). Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).**

***Variables***

The test of Hypothesis 2 (H2) is similar to the approach used for Hypothesis 1 (H1), with the key difference being the dependent variable. Instead of gender-equality attitudes, H2 focuses on left-leaning political preferences. Political attitudes are measured using the SOEP variable **plh0004**, which captures responses on an 11-point scale ranging from 0 (extreme left) to 10 (extreme right).

***Inclusion Criteria***

The analytical sample consists of parents with exactly one biological child who is at least one year old. Since the dependent variable is only available for the years 2005, 2009, 2014, and 2019, the analysis is restricted to these years. The sample includes respondents of all civil statuses—single, married, and divorced.

***Exact Analysis***

As with the test of H1, we will conduct balance diagnostics to compare parents of daughters with parents of sons. This step is necessary because the years included in the H2 analysis differ from those in H1, potentially affecting the comparability of the groups. The same set of control variables used in the H1 analysis will be applied here to ensure consistency in the evaluation of group balance.

Next, we transform the political-attitudes variable (plh0004) to align with our focus on left-leaning preferences. Since the original scale ranges from 0 (extreme left) to 10 (extreme right), we reverse the coding by subtracting each response from 10. This transformation yields a new variable, LLPP (Left-Leaning Political Preferences), where higher values indicate stronger left-leaning attitudes (e.g., an original score of 0 becomes 10, and a score of 10 becomes 0).

To test H2, we estimate the model specified in Equation 1, substituting LLPP for the original dependent variable EG.

*Interpretation*

The parameter estimate of interest is $\beta_1$, which measures differences between responders' parenting a daughter and those parenting a son in their left-leaning political preferences. Hence, $\beta_1$ measures the relative impact of having a daughter, compared to a son, on the respondents' political attitude. H2 receives support when $\hat{\beta}_1 > 0$, and p-value < .05.

**Hypothesis 3 (H3). Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters).**

*Variables*

The test of H3 differs from that of H1 in that it focuses on behaviors and outcomes that reflect gender equality, rather than on attitudes toward gender equality. Since the SOEP does not provide a single variable that directly captures gender-equal behavior, we construct a composite index to serve as the dependent variable.

This index is based on the assumption that, within a household with one biological child, gender equality is reflected in parity between parents across three key domains: power in the household, career opportunities, domestic responsibilities. We use the following variables that are indicative of behaviors or outcomes these three domains.

Power in the household:
**plc0011:** Last Word On Financial Decisions (available for years 2010, 2012, 2015, 2018)

Career opportunities:
**lm0621**: Average Working Time/Week (available in SOEP-core)

Domestic responsibilities:
**pli0043_h**: Hours Weekday Housework (available in SOEP-core)

Differences between mothers and fathers in these domains—regardless of direction—is indicative of a lack of gender equality. We operationalize this index using the following SOEP variables:

To construct the Gender Equality in Behaviors or Outcomes (GEBO) index, we first identify individual respondents who belong to the same household using household identifiers provided in the SOEP household survey. Because our focus is on gender dynamics, we exclude households with same-gender parents, as these do not allow for the analysis of gender-based differences.

For each of the three behavioral dimensions—power, opportunities, and responsibilities—we calculate the absolute difference between the responses of mothers and fathers. Smaller absolute differences indicate greater gender equality in that domain.

Assuming that all three dimensions contribute equally to the concept of gender-equal behavior, we aggregate the absolute differences across domains to construct the GEBO index. To enhance interpretability, we then normalize the index using min-max scaling, transforming it to a 0–100

scale, and then reverse scaling it. In this transformed version, a score of 0 represents the lowest observed level of gender-equal behavior, while a score of 100 reflects the highest. The GEBO index thus serves as the dependent variable in the test of Hypothesis 3 (H3).

### *Inclusion Criteria*

The analytical sample consists of heterosexual parents from the same household with exactly one biological child who is at least one year old. Since the dependent variable is only available for the years 2010, 2012, 2015, and 2018, the analysis is restricted to these years. The sample includes respondents inhabiting the same household (identified using **ppnamlpnr** and **pld0133**).

### *Exact Analysis*

As with the test of H1, we will conduct balance diagnostics to compare parents of daughters with parents of sons. This step is necessary because the years included in the H3 analysis differ from those in H1, potentially affecting the comparability of the groups. The same set of control variables used in the H1 analysis will be applied here to ensure consistency in the evaluation of group balance.

To test H3, we estimate the model specified in Equation 1, substituting GEBO for the original dependent variable EG.

### *Interpretation*

The parameter estimate of interest is $\beta_1$, which measures differences between responders' parenting a daughter and those parenting a son in their gender-equality behaviors. Hence, $\beta_1$ measures the relative impact of having a daughter, compared to a son, on the respondents' GEBO. H3 receives support when $\hat{\beta}_1 > 0$, and p-value < .05.

**Hypothesis (H4). Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)**

To test H4, we extend model (1) by adding an interaction term between the binary $Daughter_i$ variable and another describing the gender of responders. Specifically, we propose to estimate the following model:

$$EG_i = \kappa_0 + \kappa_1 Daughter_i + \kappa_2 Gender_i + \kappa_3 Daughter_i \text{x} Gender_i + \boldsymbol{W}\theta + c_y + \zeta_i \qquad (4)$$

Where $Gender_i$ is a binary variable, taking the value of one if the responder is female and zero otherwise. In favor of hypothesis 4, we will expect that $\hat{\kappa_3} < 0$. Another way to test hypothesis 4 is by estimating the predicted average values of $EG_i$ for each treatment condition (having a daughter vs. having a son) for women ($Gender_i = 1$) and men ($Gender_i = 0$) by using the marginal effects. Thus, we will have four predicted of $EG_i$, and from them we will test differences involved in Hypothesis 4.

The output from this analysis will be reported in a table like the following one:

**Table 1**. Predicted scores for $EG_i$

| **Treatment** | **Responder's gender** | **Marginal effects** |
|---|---|---|
| Daughter | Female | $m_{df}$ |
| | Male | $m_{dm}$ |
| Son | Female | $m_{sf}$ |
| | Male | $m_{sm}$ |

*Note*: $m_{lk}$ stands for the predicted marginal effect for the treatment group $l$ and the responder's gender "$k$"

From the predicted marginal effects, we will test the null hypothesis that: $Ho$: $(m_{dm} - m_{sm}) - (m_{df} - m_{sf}) = 0$, against the alternative that $Ha$: $(m_{dm} - m_{sm}) - (m_{df} - m_{sf}) > 0$. To do so, we will use an F-test.

## References

**Washington (2008)**, *"Female Socialization: How Daughters Affect Their Legislator Fathers' Voting on Women's Issues"*, *American Economic Review*. (Similar strategy to deal with endogeneity as ours).

**Oswald and Powdthavee (2010)**, *"Daughters and Left-Wing Voting"*, *Review of Economics and Statistics*. (Similar strategy to deal with endogeneity as ours. In addition, this paper studies how having a daughter affects people's political preferences).